

SCHOOL MANAGEMENT SYSTEM (SchoolMS)

PROJECT PROPOSAL DOCUMENT – PREMIUM VERSION

1. PROPOSAL SUMMARY

We propose to develop a **Premium School Management System (SchoolMS)** — a complete digital education ecosystem with enterprise web platform, dedicated mobile apps (Student + Parent + Teacher), AI-powered analytics, full LMS, ERP integration, and multi-campus management. Designed for **large school groups and educational chains (3000+ students, multi-campus)** requiring enterprise-grade scalability, automation, and performance.

Ideal For: Large school groups, CBSE/ICSE chains, residential campuses, university-linked schools (3000+ students)

2. OBJECTIVES

- Fully automate all academic and administrative operations across all campuses
- Provide a seamless digital experience for students, parents, teachers, and management
- Enable real-time multi-campus visibility with centralized control
- Implement AI-driven attendance, performance analytics, and learning insights
- Ensure enterprise-grade security, compliance, and long-term scalability

3. SCOPE OF WORK

3.1 Website + Admission Portal (Enterprise)

- Premium animated school website with SEO optimization
- Online admission portal with document upload and status tracking
- Virtual campus tour with 360-degree view
- Student and parent self-service login portal
- Alumni network and achievements portal
- Online scholarship application module

Cost: Rs. 90,000 - Rs. 1,40,000

3.2 Student Management (Full Lifecycle)

- End-to-end student lifecycle (Inquiry to Alumni)
- Digital admission with AI document verification
- QR code student ID cards

- Co-curricular, sports, and club management
- Medical records and special needs tracking
- Counseling and behavioural record management

Cost: Rs. 1,00,000 - Rs. 1,60,000

3.3 AI Face Recognition Attendance (Campus-Wide)

- Campus-wide AI face recognition for students and staff
- Multi-gate entry and exit tracking
- Real-time parent alert on entry/exit via app and WhatsApp
- Unauthorized access detection alerts
- Monthly and yearly attendance analytics with prediction

Cost: Rs. 1,20,000 - Rs. 2,00,000

3.4 Fee & Finance Management (ERP Level)

- Fully automated multi-head fee management
- Dynamic fee structure per class and category
- Online payment with multiple gateway support
- Auto installment reminders and receipts
- Scholarship, concession and sibling discount engine
- GST-compliant billing with financial year reporting
- Tally/accounting software integration (optional)

Cost: Rs. 90,000 - Rs. 1,50,000

3.5 Learning Management System (Advanced LMS)

- Chapter-wise content management (PDF/Video/Quiz)
- Live and recorded class management
- AI-powered student learning path recommendations
- Assignment submission, grading and feedback
- Student and parent progress dashboards
- Plagiarism detection for assignments

Cost: Rs. 1,00,000 - Rs. 1,60,000

3.6 Exam, Result & Academic Analytics

- Online and offline exam management
- AI-based question paper generation
- Auto result computation with multi-grading systems
- Customizable report cards with school branding
- Academic performance trend analysis per student
- Parent result delivery via app and WhatsApp

Cost: Rs. 80,000 - Rs. 1,30,000

3.7 Library, Transport & Hostel Management

- Full digital library with barcode/RFID tracking
- Real-time GPS bus tracking for parents
- Route optimization and vehicle management
- Hostel room allocation, fee and warden management
- Mess and canteen management (basic)

Cost: Rs. 80,000 - Rs. 1,30,000

3.8 Staff HRMS & Payroll

- Full HRMS with payroll automation
- Leave, shift and substitute teacher management
- Appraisal and increment management
- PF, ESI and TDS deduction automation
- Staff training and certification tracking

Cost: Rs. 80,000 - Rs. 1,30,000

3.9 Multi-Campus Management

- Centralized dashboard for all campuses
- Campus-wise performance comparison
- Centralized content and curriculum sharing
- Inter-campus student transfer management
- Consolidated finance and HR reporting

Cost: Rs. 80,000 - Rs. 1,30,000

3.10 Mobile Applications (Android / iOS)

- Student App: Timetable, homework, results, attendance, LMS, library, fees
- Parent App: Live attendance alerts, fee payment, results, bus tracking, chat
- Teacher App: Attendance marking, homework upload, marks entry, leave apply

Cost: Rs. 2,50,000 - Rs. 4,50,000

4. ADDITIONAL FEATURES

4.1 AI Academic Performance Prediction

- Early weak-student identification alerts
- Subject-wise performance trend forecasting
- Personalized remedial class recommendations

Cost: Rs. 90,000 - Rs. 1,40,000

4.2 WhatsApp + SMS + Email + Push Notifications

- Full parent communication automation

- Bulk announcement and circular delivery
- Personalized student performance alerts

Cost: Rs. 40,000 - Rs. 65,000

4.3 Online Admission CRM

- Inquiry to admission pipeline management
- Counselor-wise lead tracking
- Automated follow-up sequences
- Admission conversion analytics

Cost: Rs. 45,000 - Rs. 70,000

4.4 Security, Audit & Compliance

- Multi-level role-based access with approvals
- Full audit trail for all actions
- AES-256 data encryption
- RTE and CBSE/ICSE board compliance tools

Cost: Rs. 50,000 - Rs. 85,000

4.5 IoT Smart Campus Integration

- Smart classroom attendance via RFID
- Automated bell and PA system control
- Energy monitoring integration (optional)

Cost: Rs. 60,000 - Rs. 1,00,000

5. TECHNOLOGY STACK (MERN + AI + MOBILE)

- **Frontend:** React.js
- **Backend:** Node.js + Express.js
- **Database:** MongoDB + Redis (caching)
- **Mobile Apps:** React Native / Flutter
- **AI/ML:** Face recognition + performance prediction models
- **LMS:** Custom + Zoom/Meet API integration
- **IoT:** RFID / Arduino integration (optional)
- **Hosting:** AWS (EC2, S3, CloudFront, RDS)

6. TOTAL PROJECT COSTING

Development Cost (One-time)	Rs. 10,00,000 - Rs. 18,00,000+ (One-time)
Infrastructure	AWS Hosting: Rs. 12,000 - Rs. 55,000/month

Infrastructure	Domain + SSL: Rs. 2,000/year
Infrastructure	CDN & Backup: Rs. 18,000/year
Infrastructure	API & Integrations: Rs. 5,000 - Rs. 20,000/month
Maintenance Cost	Rs. 55,000 - Rs. 1,10,000/month/month

7. PROJECT TIMELINE

Phase	Duration
Planning	2 Weeks
Development	14-18 Weeks
Testing	3-4 Weeks
Deployment	1-2 Weeks
Total Duration	4-6 Months

8. LIMITATIONS

- High initial investment
- Requires dedicated IT team for campus-level support
- Infrastructure and third-party API costs are recurring
- Comprehensive staff training mandatory

9. FUTURE UPGRADES

- AR/VR based virtual classrooms
- Blockchain-based certificate verification
- AI chatbot for student and parent support
- Integration with CBSE/ICSE board portals
- National Academic Depository (NAD) integration

10. SUPPORT AND WARRANTY

- 3 months free support
- Priority bug fixing
- Dedicated account manager and support team
- On-site training for staff (optional)

11. TERMS AND CONDITIONS

- 40% advance payment
- 30% mid-phase
- 30% after completion
- Additional modules charged separately

12. CONCLUSION

This Premium SchoolMS solution provides a **complete digital transformation platform** for educational institutions — combining AI attendance, advanced LMS, mobile apps for students/parents/teachers, multi-campus management, smart campus IoT, and enterprise analytics, ensuring academic excellence, operational efficiency, and long-term growth.

13. CONTACT DETAILS

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